

BeatAhead — Hardcore Website Generation Prompt

Ischemic Stress Index (ISI) • Hackathon MVP

You are a senior full-stack engineer, UI/UX designer, AI product architect, and healthcare-tech product designer.

Build a complete, production-quality hackathon MVP web application called:

BEATAHEAD

Ischemic Stress Index (ISI)

AI Framework for Early Cardiac Risk Assessment

IMPORTANT:

This is a research/demo screening system, NOT a medical diagnostic device.

Never claim that the system diagnoses ischemia, heart attack, coronary artery disease, or any other medical condition.

The ISI is a screening/risk-trend indicator intended to encourage appropriate clinical follow-up and future validation.

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1. PRODUCT CONCEPT

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BeatAhead analyzes multimodal physiological signals from wearable/medical devices and produces a personalized Ischemic Stress Index (ISI) from 0-100.

Core pipeline:

PPG + ECG + SpO2 + IMU

↓

Signal Processing

↓

Artifact / Motion Removal

↓

Feature Extraction

↓

HRV + Pulse Morphology + ECG Features + SpO2 Trend

↓

AI Fusion Model

↓

Trend-Weighted ISI Engine

↓

Personalized 0-100 ISI Score

↓

Trend Visualization

↓

Screening Flag / Recommended Clinical Follow-up

The system must emphasize:

1. MULTI-SIGNAL FUSION
2. PERSONAL BASELINE
3. CONTINUOUS TREND ANALYSIS
4. EXPLAINABLE SCORE
5. NOT A DIAGNOSIS

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2. TECHNOLOGY STACK

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Use:

Frontend:

- Next.js
- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- Lucide icons
- Recharts for graphs
- Framer Motion for subtle animations

Backend:

- Next.js API routes OR FastAPI if a separate backend is necessary

For the hackathon MVP, prioritize a working frontend and realistic simulation over unnecessary infrastructure.

Do NOT introduce:

- Kubernetes

- Docker orchestration
- A2A
- MCP
- Qdrant
- PostgreSQL
- Redis
- complex microservices

unless genuinely required.

The MVP should be easy to run locally and easy to deploy.

3. DESIGN LANGUAGE

Create a premium medical-AI dashboard.

Visual direction:

- White / near-white background
- Deep navy / charcoal typography
- Restrained red accent for cardiac/risk elements
- Clean healthcare-tech aesthetic
- Modern glass/soft cards
- Thin borders
- Large whitespace
- Rounded cards
- Professional typography
- High information density without looking cluttered

Do NOT make it look like:

- a generic SaaS template
- a crypto dashboard
- a gaming dashboard
- a futuristic neon AI website

It should feel like a serious product from a medical AI startup.

Use smooth micro-interactions.

Use animations only where they communicate system activity.

4. APPLICATION STRUCTURE

Create these main routes:

```
/
/dashboard
/monitor
/signals
/trends
/insights
/about
/methodology
```

Optional:

```
/clinician
```

5. LANDING PAGE

Create a powerful landing page.

Hero:

"See the Trend Before the Event."

Subtitle:

"BeatAhead's Ischemic Stress Index fuses continuous physiological signals into a personalized risk-trend indicator designed to identify changes that may warrant clinical follow-up."

Primary CTA:

"Explore Live Monitoring"

Secondary CTA:

"How ISI Works"

Hero visual:

Create a sophisticated visualization showing:

PPG

ECG

SpO2

IMU
flowing into:
AI FUSION
then:
ISI 0-100
then:
TREND
Use animated signal lines.
Do NOT imply that the AI is directly seeing coronary blockage.

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6. PROBLEM SECTION

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Explain:
"Traditional cardiac testing is often a snapshot."
Show:
ECG
Stress Test
Angiography
as point-in-time assessments.
Then show:
CONTINUOUS PHYSIOLOGICAL TRENDS
Explain that the concept focuses on analyzing changing physiological patterns rather than isolated readings.
Also explain that existing wearable devices often expose individual metrics separately rather than combining them into a personalized ischemic-risk trend.

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7. HOW ISI WORKS

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Create a visually impressive pipeline:

STEP 01
MULTI-SENSOR INPUT
PPG
ECG
SpO2
IMU

STEP 02
SIGNAL PROCESSING
Butterworth filtering
Kalman filtering
Motion artifact rejection

STEP 03
FEATURE EXTRACTION
HRV (SDNN)
Pulse morphology
SpO2 trend
ECG-derived features

STEP 04
AI FUSION
XGBoost prototype

STEP 05
ISI SCORING
0-100 trend-weighted score

STEP 06
FOLLOW-UP
Screening flag
Clinical follow-up
Make every step interactive.
Clicking a step should open a small explanation panel.

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8. DASHBOARD

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Build the main dashboard as the centerpiece.

Header:

BeatAhead

Live Physiological Monitoring

Top cards:

CURRENT ISI

62 / 100

TREND

↑ Increasing

BASELINE

48

SIGNAL QUALITY

94%

Then create a large ISI trend graph.

X-axis:

Time

Y-axis:

ISI 0-100

Show:

- current score
- personal baseline
- previous scores
- trend line
- highlighted periods of increasing stress

IMPORTANT:

Do not simply generate random numbers.

Create a realistic simulated physiological dataset with correlated signals and a trend-weighted score.

Example:

baseline:

45-50

normal fluctuation:

40-55

temporary stress:

55-65

persistent rising trend:

65-75

Use a deterministic simulation so the dashboard looks consistent.

9. ISI SCORE VISUALIZATION

Create a large circular/donut ISI gauge.

Example:

62

ISI

"Moderate trend elevation"

Under it show:

Personal baseline: 48

Current deviation: +14

Trend: Increasing

Confidence: 86%

Add an information icon explaining:

"ISI is a research prototype screening indicator and is not a medical diagnosis."

Do NOT present arbitrary medical thresholds as clinically validated.

Instead label example ranges as:

0-30

Lower observed trend

31-60

Intermediate observed trend

61-100

Higher observed trend

And clearly state:

"Illustrative prototype ranges – not clinically validated thresholds."

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10. PERSONAL BASELINE

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This is one of the most important features.

Create a "Your Baseline" card.

Show:

Resting HR baseline

HRV baseline

SpO2 baseline

Pulse morphology baseline

Show how the current values compare against the user's own historical baseline.

Example:

HRV

Current: 42 ms

Baseline: 51 ms

Deviation: -17.6%

SpO2

Current: 96%

Baseline: 97%

Deviation: -1.0%

Explain:

"ISI is normalized against an individual's rolling baseline rather than assuming one population-wide normal value."

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11. LIVE MONITORING PAGE

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Create a live simulation interface.

Four signal panels:

PPG

ECG

SpO2

IMU

Each should have an animated real-time waveform/chart.

PPG:

Pulse waveform

ECG:

ECG-like waveform

SpO2:

continuous percentage trend

IMU:

motion/activity waveform

Controls:

[Start Monitoring]

[Pause]

[Reset Simulation]

When Start Monitoring is clicked:

- signals animate
- timestamps advance
- feature values update
- ISI changes gradually
- trend chart updates
- signal-quality indicator updates

Create a "Signal Processing" panel showing:

Raw signal

↓

Filtered signal

↓

Artifact rejection

↓

Feature extraction

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12. SIGNAL QUALITY

```
=====
Create a signal-quality system.
For every sensor display:
PPG
Signal Quality: Excellent
ECG
Signal Quality: Good
SpO2
Signal Quality: Excellent
IMU
Motion: Moderate
If motion artifacts increase:
show:
"Motion artifact detected"
and visually reduce confidence.
Do not pretend the simulated signal is real hardware data.
Clearly label the entire live demo:
"SIMULATED SENSOR DATA"
=====
```

13. FEATURES PAGE

```
=====
```

Create an interactive feature-analysis page.

Show:

HRV
SDNN
Pulse Morphology
Amplitude
Rise time
Peak characteristics
SpO2 Trend
Average
Minimum
Slope
ECG
Heart rate
RR interval
Rhythm-related prototype features
IMU
Motion intensity
Artifact detection
For each feature show:
Current
Baseline
Deviation
Trend

14. TRENDS PAGE

```
=====
```

Create long-term analytics.

Time filters:

24 HOURS
7 DAYS
30 DAYS

Graphs:

ISI trend
HRV trend
SpO2 trend
Heart rate trend
Activity / motion

Add event markers:

Rest
Exercise
High motion

Possible artifact
Create an "Insight Timeline".

Example:

08:30

Normal baseline

11:45

Increased activity

12:05

Transient ISI increase

12:30

Returned toward baseline

The wording must remain non-diagnostic.

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15. AI INSIGHTS PAGE

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Create an explainable AI interface.

Title:

"Why did the ISI change?"

Show contribution cards:

HRV contribution

Pulse morphology contribution

SpO2 trend contribution

ECG contribution

Motion/artifact contribution

Example:

HRV

Higher influence

SpO2 trend

Moderate influence

Pulse morphology

Moderate influence

Motion artifact

Low confidence impact

Use horizontal contribution bars.

Important:

Do not claim these are clinically validated feature weights.

Label them:

"Prototype model contribution visualization."

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16. RISK TREND EXPLANATION

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Instead of simply saying:

"HIGH RISK"

use language like:

"Observed physiological trend has moved above the user's recent personal baseline."

Then:

"Persistent changes may warrant discussion with a qualified healthcare professional."

Never tell the user:

"You are having a heart attack."

Never provide diagnosis.

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17. CLINICIAN VIEW

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Create an optional clinician dashboard.

Patient list:

Patient ID

Current ISI

Trend

Signal Quality

Last Updated

Click a patient.

Show:

- ISI trend
- Personal baseline
- HRV
- SpO2
- Heart rate
- Signal quality
- Feature contributions

Include:

"Clinical decision support prototype – not a diagnostic system."

Add:

Export Report

The report should contain:

- ISI trend
- baseline comparison
- signal quality
- physiological trends
- feature contribution visualization
- disclaimer

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18. DEMO MODE

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This is extremely important for the hackathon.

Create a "Demo Mode".

The website must work even without physical wearable hardware.

Provide scenarios:

- NORMAL
- STRESS EVENT
- RECOVERING
- PERSISTENT RISING TREND
- MOTION ARTIFACT

When selecting a scenario, generate corresponding simulated data.

Example:

NORMAL:

ISI $\approx$ 40-50

STRESS EVENT:

ISI gradually rises to $\approx$ 60-70 and then recovers

PERSISTENT RISING TREND:

ISI gradually increases over time

MOTION ARTIFACT:

signal quality decreases and confidence falls

Make the simulation visually convincing.

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19. AI MODEL SIMULATION

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For the MVP, create a clearly labeled prototype scoring engine.

Do NOT falsely claim a production clinical XGBoost model if none exists.

Implement a modular function such as:

calculateISI(features)

Inputs:

- HRV
- pulse morphology
- SpO2 trend
- ECG features
- motion/artifact quality
- personal baseline
- historical trend

Output:

```
{
score,
trend,
```

```
confidence,  
contributions  
}
```

The scoring engine should be deterministic.

Structure the code so that the simulated scoring function can later be replaced by a real trained XGBoost model.

Add:

```
/lib/isi/  
scoring.ts  
features.ts  
simulation.ts  
baseline.ts
```

20. DATA ARCHITECTURE

Create clean TypeScript interfaces.

Example:

PhysiologicalSample

```
{  
  timestamp,  
  ppg,  
  ecg,  
  spo2,  
  imu,  
  heartRate,  
  hrv,  
  signalQuality  
}
```

ISIScore

```
{  
  timestamp,  
  score,  
  baseline,  
  trend,  
  confidence,  
  contributions  
}
```

Do not use unstructured objects everywhere.

Keep data logic separate from UI.

21. NAVIGATION

Desktop sidebar:

```
BeatAhead  
Overview  
Live Monitor  
Signals  
Trends  
AI Insights  
Clinician View  
Methodology
```

Bottom:

```
System Status  
Demo Mode  
Settings
```

Mobile:

Use a responsive bottom navigation / hamburger menu.

22. SYSTEM STATUS

Show:

```
AI Engine  
● Online  
Sensor Stream  
● Simulated
```

Signal Processing

● Active

ISI Engine

● Active

Data Sync

● Connected

Make this clearly demo/system status, not a claim of medical-device connectivity.

23. METHODOLOGY PAGE

Create a detailed but understandable technical explanation.

Sections:

Why multimodal fusion?

Why personal baseline?

Why trends?

Signal processing

Feature engineering

AI fusion

ISI scoring

Edge AI

Cloud analytics

Clinical validation roadmap

Regulatory pathway

Clearly state:

"Clinical validation is still required."

24. ROADMAP

Show:

PHASE 1

AI Prototype

PHASE 2

Clinical Validation

PHASE 3

Regulatory Clearance

PHASE 4

Hospital & Health-Tech Integration

Also show potential future directions:

Wearable SDK

Hospital analytics

EHR integration

Health-tech licensing

25. SAFETY / MEDICAL DISCLAIMER

Put a concise disclaimer in the footer and relevant screens:

"BeatAhead ISI is a research/prototype screening indicator and is not a medical diagnosis. It is not intended to replace ECG, stress testing, angiography, physician assessment, or emergency medical care."

For demo data:

"All displayed physiological signals are simulated and do not represent a real patient's measurements."

26. RESPONSIVE DESIGN

The website must work perfectly on:

Desktop

Laptop

Tablet

Mobile

Do not just shrink the desktop UI.

Create proper responsive layouts.

27. UX QUALITY

The application should feel like a real product.

Include:

- Loading states
- Empty states
- Error states
- Tooltips
- Hover states
- Smooth transitions
- Skeleton loaders
- Toast notifications
- Accessible buttons
- Keyboard navigation
- Proper contrast
- Responsive charts

Avoid unnecessary animations.

28. CODE QUALITY

Use:

TypeScript everywhere possible.

Create reusable components.

Suggested structure:

```
/app
/dashboard
/monitor
/signals
/trends
/insights
/about
/methodology
/components
/dashboard
/charts
/signals
/isi
/layout
/ui
/lib
/isi
/simulation
/utils
```

Do not put the entire application into one huge file.

29. PERFORMANCE

Charts must remain smooth during live simulation.

Do not cause unnecessary React re-renders.

Use efficient state management.

Keep simulation intervals properly cleaned up.

Do not leak timers or event listeners.

30. FINAL REQUIREMENT

Build the actual working website, not a design mockup.

Every button that appears important must work.

Every navigation item must work.

Charts must display real generated data.

Live monitoring must actually animate.

Demo scenarios must actually change the signals and ISI.

ISI must actually update.

Baseline comparison must actually calculate.

AI contribution visualization must actually update.

The application must run with:

```
npm install  
npm run dev
```

Do not leave placeholder screens saying "Coming Soon" for core features.

Do not fabricate clinical validation.

Do not fabricate real wearable connectivity.

Use simulated data where hardware/backend integration is unavailable and label it clearly.

At the end, provide:

1. Complete project structure
2. All required files
3. Setup instructions
4. Environment variables if required
5. How the ISI simulation works
6. How to replace simulation with a real XGBoost model later
7. How to connect real wearable sensor APIs later

MOST IMPORTANT:

The final website should immediately communicate this story:

"Existing wearable signals → multimodal AI fusion → personalized baseline
→ continuous trend analysis → interpretable ISI score → appropriate
clinical follow-up."

Make the result impressive enough for a national-level hackathon demo while remaining technically honest and medically responsible.